LOCALIZING WHEN THE SITE HAS PRODUCTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The right adjoint of a cocontinuous functor. Same chapter, Lemma lemma-have-functor-other-way. Gives the description of the pushforward of the localization once a continuous right adjoint is available.
- A continuous right adjoint defines the same morphism of topoi. Same chapter, Lemma lemma-have-functor-other-way-morphism. Supplies the identification of the morphism of topoi induced by the right adjoint with the localization.
- The axioms of a site. Same chapter, Definition definition-site. Supplies the stability axiom used to check continuity.

Source reference. Original definition 00VH: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 526–526. Navigation pointer invoked at proof line 6336; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00XX: sources/stacks-sites/pinned-original-source.tex, source lines 6344–6344. Navigation pointer invoked at proof line 6344; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00XY: sources/stacks-sites/pinned-original-source.tex, source lines 6345–6345. Navigation pointer invoked at proof line 6345; the counted

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Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $U \in \mathrm{Ob}(\mathcal{C})$. Assume $\mathcal{C}$ has products of pairs of objects. Then*

1. *the functor j_U has a continuous right adjoint, namely the functor $v(X) = X \times U/U$,*
2. *the functor v defines a morphism of sites $\mathcal{C}/U \to \mathcal{C}$ whose associated morphism of topoi equals $j_U : Sh(\mathcal{C}/U) \to Sh(\mathcal{C})$, and*
3. *we have $j_{U*}\mathcal{F}(X) = \mathcal{F}(X \times U/U)$.*

Proof. The functor v being right adjoint to j_U means that given Y/U and X we have

$$\mathrm{Mor}_{\mathcal{C}}(Y, X) = \mathrm{Mor}_{\mathcal{C}/U}(Y/U, X \times U/U)$$

which is clear. To check that v is continuous let $\{X_i \to X\}$ be a covering of $\mathcal{C}$. By the third axiom of a site (Definition 00VH) we see that

$$\{X_i \times_X (X \times U) \to X \times_X (X \times U)\} = \{X_i \times U \to X \times U\}$$

is a covering of $\mathcal{C}$ also. Hence v is continuous. The other statements of the lemma follow from Lemmas 00XX and 00XY. □